Package ‘tisai’

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Title AI Enabled Time Series and Spatial Statistics

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Description Datasets used in the book ``AI enabled time series and spacial statistics''

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Roxygen list(markdown = TRUE)

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Depends R (>= 3.5.0)

Imports Matrix,glmnet,quantmod,dplyr,zoo

LazyData true

Contents

Ashare — *Selected Chinese A-Share Stock Tickers*

Description

A list of representative Chinese A-share listed companies used for constructing panel stock return data from Yahoo Finance.

Usage

```
data(Ashare)
```

Format

A data frame containing stock tickers of selected A-share companies:

`ticker` Yahoo Finance ticker symbol.

`name` Company name.

`market` Exchange code: SSE or SZSE.

Source

Yahoo Finance symbol convention for Chinese A-share listed companies: https://finance.yahoo.com/

Examples

```
data(Ashare)

## download panel data for selected A-share stocks
panel <- get_Ashare_data(
  tickers = Ashare$ticker[1:5],
  from = "2020-01-01",
  to = "2024-12-31"
)

head(panel)
```

CNgdp — *China Quarterly GDP and Sectoral Value Added*

Description

Quarterly measures of China's gross domestic product (GDP), including both current-price (nominal) and constant-price (real) series, together with value added of the primary, secondary, and tertiary industries.

Usage

```
data(CNgdp)
```

Format

A data frame with quarterly observations on:

`date` First day of the quarter (YYYY-MM-DD).

`gdp` GDP, current price (billion CNY).

`gdp1` Primary industry value added, current price (billion CNY).

`gdp2` Secondary industry value added, current price (billion CNY).

`gdp3` Tertiary industry value added, current price (billion CNY).

`rgdp` GDP, constant price (billion CNY).

`rgdp1` Primary industry value added, constant price (billion CNY).

`rgdp2` Secondary industry value added, constant price (billion CNY).

`rgdp3` Tertiary industry value added, constant price (billion CNY).

Source

National Bureau of Statistics of China (NBS), Quarterly Data Portal: https://data.stats.gov.cn/

`CNpcgdp` *Provincial Per Capita GDP in China, 2016–2024*

Description

Annual per capita GDP for 31 provincial-level regions in China from 2016 to 2024.

Usage

```
data(CNpcgdp)
```

Format

A data frame with 31 observations on the following 10 variables:

`province` Provincial-level region name.

`pcgdp2016` Per capita GDP in 2016 (CNY per person).

`pcgdp2017` Per capita GDP in 2017 (CNY per person).

`pcgdp2018` Per capita GDP in 2018 (CNY per person).

`pcgdp2019` Per capita GDP in 2019 (CNY per person).

`pcgdp2020` Per capita GDP in 2020 (CNY per person).

`pcgdp2021` Per capita GDP in 2021 (CNY per person).

`pcgdp2022` Per capita GDP in 2022 (CNY per person).

`pcgdp2023` Per capita GDP in 2023 (CNY per person).

`pcgdp2024` Per capita GDP in 2024 (CNY per person).

Source

National Bureau of Statistics of China (NBS), data portal: https://data.stats.gov.cn/

CNpower *China Monthly Electricity Generation by Source*

Description

Monthly electricity generation in China, including total generation and generation by thermal, hydro, nuclear, wind, and solar sources.

Usage

```
data(CNpower)
```

Format

A data frame with monthly observations on:

`date` First day of the month (YYYY-MM-DD).

`power` Total electricity generation (100 million kWh).

`thermal` Thermal power generation (100 million kWh).

`hydro` Hydropower generation (100 million kWh).

`nuclear` Nuclear power generation (100 million kWh).

`wind` Wind power generation (100 million kWh).

`solar` Solar power generation (100 million kWh).

Source

National Bureau of Statistics of China (NBS), Monthly Data Portal: https://data.stats.gov.cn/

CNprice *China Monthly Consumer Price Indices (YoY)*

Description

Monthly consumer price indices (CPI), reported as year-over-year indices (same month of previous year = 100), for the national aggregate as well as urban and rural residents.

Usage

```
data(CNprice)
```

Format

A data frame with monthly observations on:

`date` First day of the month (YYYY-MM-DD).

`cpi` National CPI (YoY index, same month last year = 100).

`cpi_city` Urban CPI (YoY index, same month last year = 100).

`cpi_rural` Rural CPI (YoY index, same month last year = 100).

Source

National Bureau of Statistics of China (NBS), Monthly Data Portal: https://data.stats.gov.cn/

`Energy` *Selected Global Energy Indicators*

Description

Annual energy-related indicators for selected countries.

Usage

```
data(Energy)
```

Format

A data frame containing:

`country` Country name.

`year` Year.

`pop` Population.

`gdp` Gross domestic product.

`energy` Primary energy consumption.

`elec` Electricity generation.

`renew` Share of renewable energy.

Source

Our World in Data energy dataset. https://github.com/owid/energy-data

---

`get_CNmarket_data` *Download Chinese financial market data from Yahoo Finance*

Description

This function downloads daily adjusted closing prices for major Chinese financial market indices and related financial variables from Yahoo Finance, aligns trading dates across markets, and returns a data frame.

Usage

```
get_CNmarket_data(from = "2018-01-01", to = "2025-12-31", save_path = NULL)
```

Arguments

`from`	A character string specifying the start date in `"YYYY-MM-DD"` format. The default is `"2018-01-01"`.
`to`	A character string specifying the end date in `"YYYY-MM-DD"` format. The default is `"2025-12-31"`.
`save_path`	An optional character string giving the file path to save the resulting data object as an `.rda` file. If `NULL`, the data are not saved to disk.

Details

The default series include the Shanghai Composite Index, Shenzhen Component Index, CSI 300 Index, Hang Seng Index, and the USD/CNY exchange rate.

The function retrieves market data from Yahoo Finance via `quantmod::getSymbols()`. Since different markets may have different trading calendars, the downloaded series are aligned by date using inner joins, so that only common trading days are retained.

The default symbols are:

shanghai Shanghai Composite Index (`000001.SS`)

shenzhen Shenzhen Component Index (`399001.SZ`)

hs300 CSI 300 Index (`000300.SS`)

hsi Hang Seng Index (`^HSI`)

usd_cny USD/CNY exchange rate (`CNY=X`)

Value

A data frame containing aligned daily adjusted closing prices for the selected Chinese and related financial market series. The first column is `date`, followed by the downloaded market variables.

Examples

```
## Not run:
library(tisai)

CNmarket <- get_CNmarket_data()
head(CNmarket)

CNmarket2 <- get_CNmarket_data(
  from = "2020-01-01",
  to = "2024-12-31"
)

CNmarket3 <- get_CNmarket_data(
  save_path = "CNmarket.rda"
)

## End(Not run)
```

get_USmarket_data *Download global financial market data from Yahoo Finance*

Description

This function downloads daily adjusted closing prices for major global financial market assets from Yahoo Finance, aligns trading dates across markets, and returns a data frame.

Usage

```
get_USmarket_data(
  from = "2018-01-01",
  to = "2025-12-31",
  symbols = NULL,
  save_path = NULL
)
```

Arguments

`from`	A character string specifying the start date in `"YYYY-MM-DD"` format. The default is `"2018-01-01"`.
`to`	A character string specifying the end date in `"YYYY-MM-DD"` format. The default is `"2025-12-31"`.
`symbols`	An optional named character vector of Yahoo Finance symbols. The names of this vector will be used as the output variable names. If `NULL`, a default set of symbols is used.
`save_path`	An optional character string giving the file path to save the resulting data object as an `.rda` file. If `NULL`, the data are not saved to disk.

Details

The default series include the S\&P 500 index, Nasdaq Composite index, U.S. dollar index, and gold futures price.

The function retrieves market data from Yahoo Finance via `quantmod::getSymbols()`. Since different markets may have different trading calendars, the downloaded series are aligned by date using inner joins, so that only common trading days are retained.

The default symbols are:

sp500 S\&P 500 Index (`^GSPC`)

nasdaq Nasdaq Composite Index (`^IXIC`)

dxy U.S. Dollar Index (`DX-Y.NYB`)

gold Gold Futures (`GC=F`)

Value

A data frame containing aligned daily adjusted closing prices for the selected market series. The first column is `date`, followed by the downloaded market variables.

Examples

```
## Not run:
library(tisai)

USmarket <- get_USmarket_data()
head(USmarket)

USmarket2 <- get_USmarket_data(
  from = "2020-01-01",
  to = "2024-12-31"
)

USmarket3 <- get_USmarket_data(
  save_path = "USmarket.rda"
)

## End(Not run)
```

Gtemp *Global Annual Temperature Anomalies (1850–2017)*

Description

This dataset provides the global annual mean temperature anomalies relative to the 1961–1990 average, sourced from the HadCRUT4 dataset.

Usage

```
data(Gtemp)
```

Format

A data frame with 168 observations on the following 2 variables:

`year` The year of observation (from 1850 to 2017).

`anomaly` Global temperature anomaly (degrees Celsius) relative to the 1961–1990 mean.

Details

The dataset reflects the long-term warming trend and potential 60-70 year periodic oscillations in global temperatures.

Source

Climatic Research Unit (CRU), University of East Anglia (HadCRUT4). https://crudata.uea.ac.uk/cru/data/temperature/

References

Wang, S., You, J., & Huang, T. (2022). Modelling and applications for non-stationary time series in the presence of trend and period. Scientia Sinica Mathematica, 52(2), 177-208.

Examples

```
data(Gtemp)
plot(Gtemp$year, Gtemp$anomaly, type = "l", main = "Global Temperature Anomalies")
```

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gtemp.month *Global Temperature Monthly Data*

Description

Global temperature monthly data from the book "AI enabled time series and spacial statistics"

Usage

```
data(gtemp.month)
```

Format

A data frame with 12 rows (months) and 49 columns (years from 1975 to 2023). Each cell contains the average temperature for that month and year.

Details

Global temperature monthly data

This dataset contains global temperature monthly data from the book "AI enabled time series and spacial statistics".

The gtemp.month dataset provides monthly global temperature data from 1975 to 2023. Rows represent months (1-12) and columns represent years. This dataset is often used in climate change analysis and time series modeling.

Source

Package 'astsa' available at https://nickpoison.github.io/

Examples

```
data(gtemp.month)

# Transpose the data for plotting
gtemp_t <- t(gtemp.month)

# Plot the temperature data for January
plot(rownames(gtemp_t), gtemp_t[, 1], type = "l",
     main = "January Global Temperature",
     xlab = "Year", ylab = "Temperature",
     col = "blue")

# Calculate summary statistics for each month
apply(gtemp.month, 1, summary)
```

Hare *Hare Population Data*

Description

Hare population data from the book "AI enabled time series and spacial statistics"

Usage

```
data(Hare)
```

Format

A time series object with the following characteristics:

- Time period: 1845 - 1935
- Frequency: Annual
- Values: Hare population counts

Details

Hare population data

This dataset contains hare population data from the book "AI enabled time series and spacial statistics".

The Hare dataset provides annual population counts of hares from 1845 to 1935. This dataset is often used in time series analysis and population dynamics studies.

Source

Package 'astsa' available at https://nickpoison.github.io/

Examples

```
data(Hare)

# Plot the hare population data
plot(Hare, main = "Hare Population Data",
     xlab = "Year", ylab = "Population",
     col = "blue")

# Calculate summary statistics
summary(Hare)
```

HKflu *Weekly Influenza Surveillance and Environmental Data in Hong Kong*

Description

This dataset contains weekly records of influenza incidence rates along with various air quality and meteorological indicators in Hong Kong from December 30, 2012, to December 22, 2019.

Usage

```
data(HKflu)
```

Format

A data frame with 365 observations on the following 11 variables:

- `week` A numeric vector indicating the sequence of monitored weeks.
- `influenza` Influenza intensity, measured as the number of laboratory-confirmed influenza cases per 1,000 recorded consultations.
- `FSP` Fine Suspended Particulates (PM2.5) daily average concentration.
- `RSP` Respirable Suspended Particulates (PM10) daily average concentration.
- `NO2` Nitrogen Dioxide (NO_2) daily average concentration.
- `CO` Carbon Monoxide (CO) daily average concentration.
- `SO2` Sulphur Dioxide (SO_2) daily average concentration.
- `NOX` Nitrogen Oxides (NO_x) daily average concentration.
- `O3` Ozone (O_3) daily average concentration.
- `temperature` Weekly average temperature in degrees Celsius.
- `humidity` Weekly average relative humidity percentage.

Source

Hong Kong Environmental Protection Department and relevant health authorities.

References

Li, J., Wang, S., & You, J. (2022). Nonparametric additive models for discrete-time series with periodic features. Acta Mathematica Sinica, Chinese Series, 65(1), 177-204.

Examples

```
data(HKflu)
summary(HKflu)
plot(HKflu$temperature, HKflu$influenza)
```

HousePrice — *Monthly House Price Index for Selected U.S. Cities*

Description

Monthly Zillow Home Value Index (ZHVI) for selected major U.S. cities.

Usage

```
data(HousePrice)
```

Format

A data frame containing:

`region` City name.

`state` State name.

`date` Month.

`price` Typical home value (USD).

Source

Zillow Research Housing Data. https://www.zillow.com/research/data/

lap — *Local Area Pollution (LAP) Data*

Description

LAP data from the book "AI enabled time series and spacial statistics"

Usage

```
data(lap)
```

Format

A multivariate time series (mts) object with the following characteristics:

- Time period: 1970 - 1980
- Frequency: Weekly (52 observations per year)
- Number of variables: 11
- Variables:
  - tmort: Total mortality
  - rmort: Respiratory mortality
  - cmort: Cardiovascular mortality
  - tempr: Temperature
  - rh: Relative humidity

- co: Carbon monoxide
- so2: Sulfur dioxide
- no2: Nitrogen dioxide
- hycarb: Hydrocarbons
- o3: Ozone
- part: Particulate matter

Details

LAP data

This dataset contains LAP (Local Area Pollution) data from the book "AI enabled time series and spacial statistics".

The lap dataset provides weekly measurements of various pollution and health indicators from 1970 to 1980. This dataset is often used in environmental health studies and time series analysis of pollution effects.

Source

Package 'astsa' available at https://nickpoison.github.io/

Examples

```
data(lap)

# Plot the first variable (total mortality)
plot(lap[, 1], main = "Total Mortality",
     xlab = "Time", ylab = "Mortality Rate",
     col = "red")

# Calculate correlation between temperature and ozone
cor(lap[, "tempr"], lap[, "o3"], use = "complete.obs")

# Plot multiple variables
plot(lap[, c("tempr", "o3", "co")],
     main = "Temperature, Ozone, and Carbon Monoxide")
```

Lynx *Lynx Population Data*

Description

Lynx population data from the book "AI enabled time series and spacial statistics"

Usage

```
data(Lynx)
```

Format

A time series object with the following characteristics:

- Time period: 1845 - 1935
- Frequency: Annual
- Values: Lynx population counts

Details

Lynx population data

This dataset contains lynx population data from the book "AI enabled time series and spacial statistics".

The Lynx dataset provides annual population counts of lynx from 1845 to 1935. This dataset is often used in time series analysis and predator-prey relationship studies.

Source

Package 'astsa' available at https://nickpoison.github.io/

Examples

```
data(Lynx)

# Plot the lynx population data
plot(Lynx, main = "Lynx Population Data",
     xlab = "Year", ylab = "Population",
     col = "red")

# Calculate summary statistics
summary(Lynx)
```

Power *Hourly Household Electricity Consumption*

Description

Hourly electricity consumption of a household in 2010.

Usage

```
data(Power)
```

Format

A data frame containing:

`time` Hourly timestamp

`power` Active power (kilowatt)

`reactive` Reactive power (kilowatt)

`voltage` Voltage (volt)

`current` Current intensity (ampere)

`sub1` Energy sub-metering 1

`sub2` Energy sub-metering 2

`sub3` Energy sub-metering 3

Source

UCI Machine Learning Repository. https://archive.ics.uci.edu/ml/datasets/individual+household+electric+power+consumption

USeconomic	*U.S. Quarterly National Economic Output and Expenditure Components*

Description

This dataset provides a structural (quarterly) snapshot of the U.S. economy from 2000.1.1 to 2025.7.1, consisting of 103 observations.

It integrates the major components of Gross Domestic Product (GDP): - Aggregate Output: Nominal and Real GDP. - Domestic Demand: Personal consumption (PCE) and Private investment. - External Sector: Real exports and imports of goods and services.

The dataset allows students to explore the nonlinear dynamics of investment shocks and the global trade balance using advanced AI structural models.

Usage

```
data(USeconomic)
```

Format

A data frame with aligned quarterly observations on the following 7 variables:

`date` Start date of the quarter (YYYY-MM-DD).

`gdp` Gross Domestic Product (Billions of Dollars).

`real_gdp` Real Gross Domestic Product (Billions of Chained 2017 Dollars).

`pce` Personal Consumption Expenditures (Billions of Dollars).

`invest` Real Gross Private Domestic Investment (Billions of Chained 2017 Dollars).

`export` Real Exports of Goods and Services (Billions of Chained 2017 Dollars).

`import` Real Imports of Goods and Services (Billions of Chained 2017 Dollars).

Source

U.S. Bureau of Economic Analysis (BEA) via Federal Reserve Bank of St. Louis (FRED). Data IDs: GDP, GDPC1, PCE, GPDIC1, EXPGSC1, and IMPGSC1. Visit: https://fred.stlouisfed.org/

References

U.S. Bureau of Economic Analysis, "National Income and Product Accounts". Retrieved from FRED: https://fred.stlouisfed.org/categories/106

Examples

```
data(USeconomic)
# Calculate Investment-to-GDP ratio
inv_ratio <- USeconomic$invest / USeconomic$real_gdp
plot(USeconomic$date, inv_ratio, type="l", ylab="Investment/GDP Ratio")
```

`USmacro` *U.S. Monthly Macroeconomic Pulse: Labor, Price, and Monetary Indicators*

Description

This dataset provides a high-frequency (monthly) snapshot of the U.S. economy from 2000.1.1 to 2025.10.1, consisting of 310 observations.

The variables cover four critical dimensions: 1. Labor Market: Unemployment rate and Nonfarm payrolls. 2. Price Level: Consumer Price Index (CPI) for inflation tracking. 3. Monetary Policy: Federal Funds Rate and 10-Year Treasury yields. 4. Real Activity: Industrial production and Housing starts.

This data is ideal for teaching lead-lag relationships, regime switching, and high-frequency forecasting using LSTM or Transformer architectures.

Usage

```
data(USmacro)
```

Format

A data frame with aligned monthly observations on the following 8 variables:

`date` First day of the month (YYYY-MM-DD).

`unrate` Civilian Unemployment Rate (Percent).

`cpi` Consumer Price Index for All Urban Consumers (Index 1982-1984=100).

`interest` Federal Funds Effective Rate (Percent).

`indpro` Industrial Production Index (Index 2017=100).

`employ` All Employees, Total Nonfarm (Thousands of Persons).

`housing` New Privately Owned Housing Units Started (Thousands of Units).

`bond10y` 10-Year Treasury Constant Maturity Rate (Percent).

Source

Federal Reserve Bank of St. Louis (FRED). Data accessible via the following IDs: UNRATE, CPIAUCSL, FEDFUNDS, INDPRO, PAYEMS, HOUST, and GS10. Visit: https://fred.stlouisfed.org/

References

Federal Reserve Bank of St. Louis, Federal Reserve Economic Data (FRED). https://fred.stlouisfed.org

Examples

```
data(USmacro)
# Visualize the relationship between interest rates and housing starts
plot(USmacro$interest, USmacro$housing,
     main="Interest Rates vs Housing Starts")
```

`WorldMacro` *Global Macroeconomic Panel Data*

Description

Panel data of macroeconomic indicators for multiple countries, including GDP per capita, population, and inflation rate.

Usage

```
data(WorldMacro)
```

Format

A data frame containing:

`country` Country name.

`year` Year.

`gdp_pc` GDP per capita (constant USD).

`population` Total population.

`inflation` Inflation rate.

Source

World Bank World Development Indicators https://data.worldbank.org

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